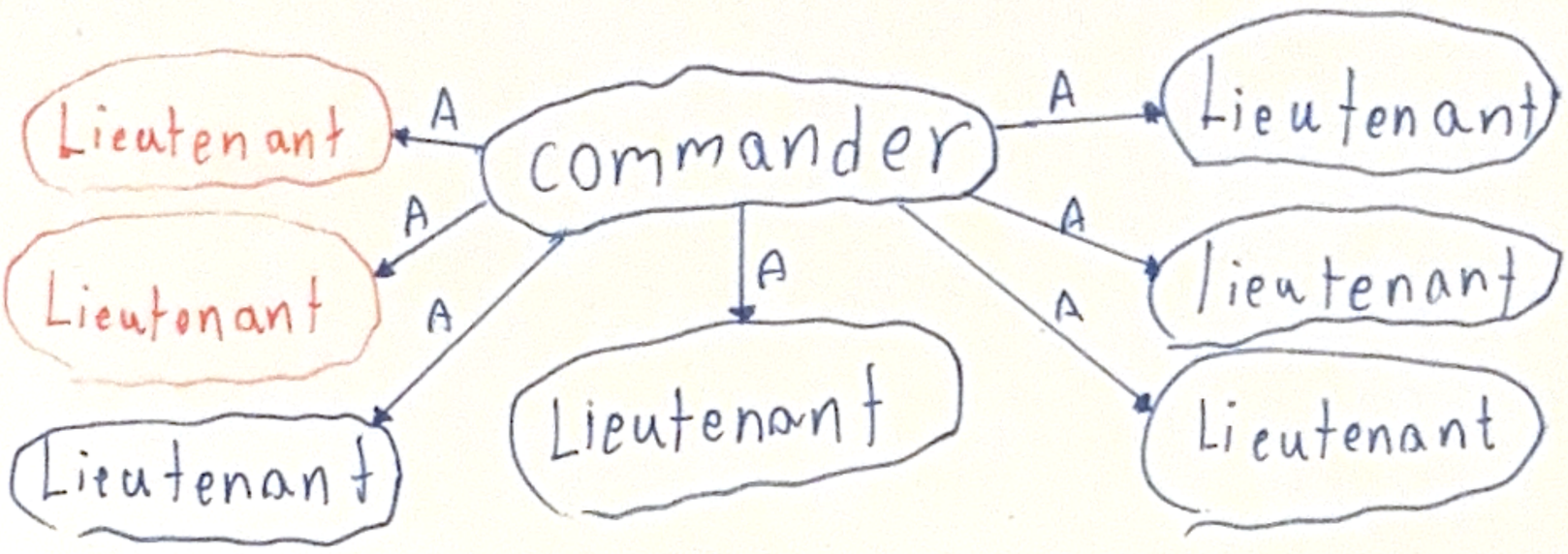
Exercise: Byzantine Fault Tolerance

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1. you need to briefly describe the process of Oral Message algorithm in your example.



Sol

This scenario $N = 8$ nodes total

and $f = 2$

This satisfies the Byzantine condition $N > 3f$

$8 > 3(2) \Rightarrow 8 > 6$ True

Phase 1 The loyal commander sends A (Attack) directly to all 7 Lieutenant

Phase 2 Each of the Lieutenant now relay the value they received to the other 6.

- the 5 blue relay A to all other.
- the 2 red relay R to all other

Phase 3 Each loyal Lieutanant all receive messages and decide by majority rule.

commander ↕ { A, A, A, A, A, R, R } loyal L, traitor L

for example for L1 would receive

majority is "Attack" (5 > 2)

All 5 loyal lieutanant will arrive at the same tally and will "Attack" and overrule the 2 "Retreat" successfully.

∴ The lieutenant successfully follow the order from the commander